

Keshav Krishna

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PROFESSIONAL SUMMARY

MS Computer Science candidate at NYU (May 2027) with 2+ years of industry ML/data science experience. Data Scientist at JioSaavn (12M+ daily users) shipping recommendation systems, RAG pipelines, and LLM-based personalization; Risk Technology Intern at Moore Capital; Computer Vision Intern at GE Healthcare. Hands-on with LLM post-training (SFT/GRPO), GPU kernels (FlashAttention-2 in Triton), and end-to-end ML systems. Seeking 2027 new-grad ML, data science, and software engineering roles.

EDUCATION

New York University Aug 2025 - May 2027
MS, Computer Science GPA: 3.95/4.0

Indian Institute of Technology Ropar Aug 2019 - Jun 2023
BTech, Computer Science and Engineering GPA: 8.74/10

SKILLS

Programming Languages: Python, C/C++, SQL, JavaScript, Bash

Machine Learning & GenAI: PyTorch, Hugging Face, transformers, LLMs, RAG, SFT/DPO/GRPO/RLHF, quantization (4-bit NF4, bitsandbytes), QLoRA, vLLM, inference optimization, evaluation & benchmarking, agentic systems, MCP

Data & Infrastructure: Spark, Hive, Airflow, Docker, Kubernetes, Triton, NVIDIA Nsight Systems, Slurm/HPC, multi-GPU serving, GCP/Azure

Databases & Web: Postgres, MySQL, Redis, Qdrant, FastAPI, Flask, REST APIs, Git, Linux

WORK EXPERIENCE

Moore Capital Management | Risk Technology Intern May 2026 - Present

- Owned risk tooling end to end: Python data workflows behind HTML/CSS dashboards, taken from a verbal stakeholder request to production in days and now used daily by 20+ risk, portfolio and operations staff.
- Optimized VaR, CVaR, beta and correlation calculations with vectorized Python and Numba JIT, cutting runtime about 70% on multi-million-row datasets so daily risk numbers are ready before market open.
- Benchmarked distributed data-processing tools on the firm's compute cluster with profiling and system modeling, locating scalability bottlenecks that guided infrastructure decisions and lifting batch throughput about 2x.

JioSaavn | Data Scientist Jul 2023 - Jul 2025

- Built early-morning and late-night content mixes for 12M daily users using Hive, Spark, Redis, and PHP, driving 1M impressions, 800K clicks, and 500K streams per day.
- Generated mood-based topic mixes for 7.6M daily users using Hive, Spark, and TF-IDF; added deduplication filters and Flask endpoints to the PHP backend; served the feature to 100M app users, achieving 12.81% CTR and 8.5 streams per user.
- Assessed few-shot SetFit and transformer models including HingRoBERTa and mURIL-base, then transitioned to LLaMA3-8B-Instruct prompt engineering, achieving 85% accuracy on manually labeled playlist-moderation data and absorbing about 60% of the manual review load.
- Quantized the LLaMA3-8B-Instruct moderation model to 4-bit with bitsandbytes (NF4, QLoRA-style) and served it with multi-GPU batched inference, cutting the weight footprint roughly 4x versus FP16 so the moderation workload fit on the existing inference GPUs while holding the 85% accuracy bar.
- Created a custom BERT model for song and user embeddings, improving top-5 and top-10 recommendation accuracy to 61.82% and 67.21% and strengthening user-learning/personalization capabilities.
- Developed an LLaMA3-8B-Instruct RAG system for prompt intent classification that retrieved songs dynamically to assemble playlists and generated custom cover art using the segmind/SSD-1B diffusion model. Owned deployment and inference optimization on Docker and Kubernetes, serving 50K+ generated playlists per day at sub-2s p95 latency.
- Reduced infrastructure cost and operational debt by migrating data from Annoy indices to Qdrant, purging unused data, and cleaning stale tables for improved lifecycle management.

GE Healthcare | Computer Vision Intern May 2022 - Jul 2022

- Built YOLO-based medical-image de-identification and RGB/IR people-detection pipelines over 20,000+ images, reaching 95% detection accuracy and cutting manual image review about 80%.
- Packaged reusable computer-vision modules as Docker images and exposed functionality through FastAPI endpoints for easier integration across projects.

PROJECTS

FlashAttention-2: From-Scratch Triton Implementation with GPU Profiling

- Hand-wrote the FlashAttention-2 forward and backward kernels in OpenAI Triton: 2D launch grid over (query tiles, batch), block-pointer tiled loads, online softmax with running max and logsumexp, compile-time causal masking, and FP32 accumulation buffers for mixed-precision inputs.
- Implemented the recomputation-based backward pass that rebuilds $P = \exp(S - L)$ from saved Q, K, V and L instead of materialising the $N \times N$ attention matrix, and pre-computes $D = \text{rowsum}(O * dO)$; this cut attention memory from $O(N^2)$ to $O(N)$ and took 65K-token attention from 10+ GB to under 1 GB, making long-context training feasible.
- Built the surrounding benchmarking and profiling harness: sweep over sequence lengths 128 to 65,536, five model sizes up to 2.7B parameters, FP32 vs BF16 mixed precision via torch.autocast, NVTX-annotated NVIDIA Nsight Systems traces, PyTorch CUDA memory snapshots and OOM accounting; measured about 2x speedup over naive attention and compared against torch.compile.

Alignment and Reasoning RL for Mathematical LLMs

- Built an end-to-end post-training pipeline for reasoning models: zero-shot evaluation, SFT, GRPO with verified rewards, masked losses, entropy tracking, policy-gradient updates, and automatic preference-pair generation for DPO/GRPO runs.
- Fine-tuned Qwen2.5-Math-1.5B on chain-of-thought data through SFT then GRPO, lifting MATH accuracy from 56.4% to 60.4% (+4.0 points), with vLLM serving the rollouts and evaluation passes.

Transformer Language Model from Scratch

- Implemented the model without torch.nn high-level modules: Linear, Embedding, RMSNorm, numerically stable softmax, causal multi-head self-attention, SwiGLU feed-forward, RoPE, cross-entropy loss, AdamW, cosine LR schedule with warmup, gradient clipping and checkpointing; reached validation loss 3.1 on TinyStories.
- Wrote a byte-level BPE tokenizer from scratch with multiprocessing parallel pre-tokenization, incremental merge counting (only pairs adjacent to a merge are re-counted) and a streaming encode_iterable for files that do not fit in memory; memory-mapped loading cut data-pipeline time about 3x.
- Ran systematic architecture ablations (RMSNorm vs none, pre- vs post-norm, RoPE vs NoPE, SwiGLU vs SiLU) with W&B sweep tracking; pre-norm + RoPE + SwiGLU won by about 8% validation loss, and analytical FLOPs, activation-memory and MFU accounting covered GPT-2 Small through XL configurations.

Phase-Adaptive Generation for Diffusion Language Models (NYU research, Prof. Greg Durrett)

- Analyzed the generation dynamics of Dream/LLaDA-style diffusion language models through per-token refinement traces, revealing interpretable phase structure: template spans stabilize immediately while arithmetic and answer-finalization spans demand substantially more refinement.
- Trained Transformer predictors over 138K block-level control tuples to forecast per-block refinement budgets, and integrated them into the LLaDA decoding loop, matching AdaBlock's 89.5% GSM8K accuracy with about 21% fewer forward evaluations.

ML-Driven Program Phase Prediction for Dynamic Hardware Resource Optimization (NYU, Prof. Mohamed Zahran)

- Clustered temporal traces of multicore workloads into interpretable pressure states and distilled them into compact predictors with sub-KB model state, reaching 90%+ phase-prediction accuracy while keeping per-timestep overhead negligible.
- Framed the predictor as a runtime scheduling input and evaluated it against hardware-counter baselines with ablation studies over phase granularity and feature sets.

PUBLICATIONS

- Advancements in Cache Management: A Review of Machine Learning Innovations — Frontiers in Artificial Intelligence, 2025. Single-author review of 40+ papers on ML for cache management. <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1441250/full>
- LLC Intra-set Write Balancing for Non-Volatile Memory Caches — arXiv, 2024. Bachelor's thesis research on non-volatile memory caches. <https://arxiv.org/abs/2410.15344>